

Cointeracting bialgebras and applications to graphs

Loïc Foissy

FoCM 2026–Wien

Let G a (proto)-algebraic monoid. The algebra $\mathbb{C}[G]$ of polynomial functions on G inherits a coproduct $\Delta : \mathbb{C}[G] \longrightarrow \mathbb{C}[G] \otimes \mathbb{C}[G] \approx \mathbb{C}[G \times G]$ such that

$$\forall f \in \mathbb{C}[G], \forall x, y \in G, \quad \Delta(f)(x, y) = f(xy).$$

This makes $\mathbb{C}[G]$ a bialgebra. It is a Hopf algebra if, and only if, G is a group. Moreover, G is isomorphic to the monoid $\text{Char}(\mathbb{C}[G])$ of characters of $\mathbb{C}[G]$.

Characters of a bialgebra B

A character of a bialgebra B is an algebra morphism $\lambda : B \longrightarrow \mathbb{C}$. The set of characters $\text{Char}(B)$ is given an associative convolution product:

$$\lambda * \mu = m_{\mathbb{C}} \circ (\lambda \otimes \mu) \circ \Delta.$$

Let G and G' be two (proto)-algebraic monoids, such that G' acts on G by monoid endomorphisms (on the right).

Cointeracting bialgebras

- 1 $A = (\mathbb{C}[G], m_A, \Delta)$ is a bialgebra.
- 2 $B = (\mathbb{C}[G'], m_B, \delta)$ is a bialgebra.
- 3 B coacts on A by a coaction $\rho : A \longrightarrow A \otimes B$.
- 4 A is a bialgebra in the category of B -comodules.

In other words, for any $f, g \in A$,

$$\rho(fg) = \rho(f)\rho(g),$$

$$\rho(1_A) = 1_A \otimes 1_B,$$

$$(\varepsilon_A \otimes \text{Id}_B) \circ \rho(f) = \varepsilon_A(f)1_B,$$

$$(\Delta \otimes \text{Id}) \circ \rho(f) = m_{1,3,24} \circ (\rho \otimes \rho) \circ \Delta(f).$$

Polynomials

The group $(\mathbb{C}^*, \times)$ acts on $(\mathbb{C}, +)$ by group automorphisms.

- $A = (\mathbb{C}[X], m, \Delta)$ with $\Delta(X) = X \otimes 1 + 1 \otimes X$.
- $B = (\mathbb{C}[X, X^{-1}], m, \delta)$ with $\delta(X) = X \otimes X$.
- $\rho(X) = X \otimes X$.

Polynomials

The monoid $(\mathbb{C}, \times)$ acts on $(\mathbb{C}, +)$ by group endomorphisms.

- $A = (\mathbb{C}[X], m, \Delta)$ with $\Delta(X) = X \otimes 1 + 1 \otimes X$.
- $B = (\mathbb{C}[X], m, \delta)$ with $\delta(X) = X \otimes X$.
- $\rho = \delta$.

From now, we shall consider only examples where $A = B$ as algebras: we obtain objects (A, m, Δ, δ) , with one product and two coproducts. The coaction ρ and the coproduct δ are equal. These objects will be called double bialgebras.

Definition

A double bialgebra is a family (A, m, Δ, δ) such that:

- ① (A, m, δ) is a bialgebra.
- ② (A, m, Δ) is a bialgebra in the category of comodules over (A, m, δ) , with the coaction $\rho = \delta$.

In other words, (A, m, Δ) is a bialgebra, with the compatibilities

$$\begin{aligned}
 (\varepsilon_{\Delta} \otimes \text{Id}) \circ \delta(x) &= \varepsilon_{\Delta}(x) 1, \\
 (\Delta \otimes \text{Id}) \circ \delta &= m_{1,3,24} \circ (\delta \otimes \delta) \circ \Delta.
 \end{aligned}$$

Polynomials

$(A, m) = (B, m) = (\mathbb{C}[X], m)$, where m is the usual product of polynomials, and the coproducts are given by

$$\Delta(X^n) = \sum_{k=0}^n \binom{n}{k} X^k \otimes X^{n-k},$$
$$\delta(X^n) = X^n \otimes X^n.$$

$\mathcal{H}_{\mathcal{G}}$ has for basis the set of graphs:

$$1; \cdot; \vdash, \dots; \nabla, \vee, \vdash, \dots;$$

$$\boxtimes, \boxdot, \boxplus, \boxminus, \boxtimes, \boxdot, \boxplus, \boxminus, \nabla \cdot, \vee \cdot, \vdash \vdash, \vdash \dots, \dots$$

The product is the disjoint union m .

The unit is the empty graph 1 .

(Schmitt, 1994). The first coproduct Δ is defined by

$$\Delta(G) = \sum_{V(G)=I \sqcup J} G|_I \otimes G|_J.$$

Examples

$$\Delta(\nabla) = \nabla \otimes 1 + 1 \otimes \nabla + 3! \otimes . + 3. \otimes !,$$

$$\Delta(\vee) = \vee \otimes 1 + 1 \otimes \vee + 2! \otimes . + .. \otimes . + 2. \otimes ! + . \otimes \dots$$

The counit is given by

$$\varepsilon_{\Delta}(G) = \delta_{G,1}.$$

(Schmitt, 1994 – Manchon, 2011). The second coproduct δ is defined by

$$\delta(G) = \sum_{\sim} (G / \sim) \otimes (G | \sim),$$

where

- $\sim$ runs in the set of equivalences on $V(G)$ which classes are connected.
- $G | \sim$ is the union of restrictions of G to a class of $\sim$.
- $G / \sim$ is obtained by the contraction of the equivalence classes of $\sim$.

Examples

$$\delta(\cdot) = \cdot \otimes \cdot,$$

$$\delta(!) = \cdot \otimes ! + ! \otimes \cdot,$$

$$\delta(\nabla) = \cdot \otimes \nabla + 3! \otimes \cdot + \nabla \otimes \cdot,$$

$$\delta(\vee) = \cdot \otimes \vee + 2! \otimes \cdot + \vee \otimes \cdot.$$

Its counit is given by

$$\epsilon_\delta(G) = \begin{cases} 1 & \text{if } G \text{ has no edge,} \\ 0 & \text{otherwise.} \end{cases}$$

Questions

- Theoretical consequences?
- Combinatorial applications?
- Other examples?

We consider a double bialgebra (B, m, Δ, δ) .

Proposition

Let (B', m', Δ') be a bialgebra and let $E_{B \rightarrow B'}$ be the set of bialgebra morphisms from (B, m, Δ) to (B', m', Δ') . The monoid of characters $\text{Char}(B)$ of (B, m, δ) acts on $E_{B \rightarrow B'}$ by

$$\phi \curvearrowright \lambda = (\phi \otimes \lambda) \circ \delta.$$

When $B' = \mathbb{C}[X]$, under a condition of connectedness on the coproduct Δ :

Theorem

- 1 There exists a unique morphism $\phi_1 : B \longrightarrow \mathbb{C}[X]$, compatible with both bialgebraic structures.
- 2 The following maps are bijections, inverse one from the other:

$$\left\{ \begin{array}{ccc} \text{Char}(B) & \longrightarrow & E_{B \rightarrow \mathbb{C}[X]} \\ \lambda & \longmapsto & \phi_1 \leftarrow \lambda, \end{array} \right. \quad \left\{ \begin{array}{ccc} E_{B \rightarrow \mathbb{C}[X]} & \longrightarrow & \text{Char}(B) \\ \phi & \longmapsto & \epsilon_\delta \circ \phi \\ & & = \phi(\cdot)(1). \end{array} \right.$$

Proposition

If $\lambda \in \text{Char}(B)$, for any $x \in B$, with $\varepsilon_\Delta(x) = 0$,

$$\phi_1 \rightsquigarrow \lambda(x) = \sum_{n=1}^{\infty} \lambda^{\otimes n} \circ \tilde{\Delta}^{(n-1)}(x) \frac{X(X-1) \cdots (X-n+1)}{n!}.$$

Here, $\tilde{\Delta}$ is the reduced coproduct

$$\tilde{\Delta}(x) = \Delta(x) - x \otimes 1 - 1 \otimes x,$$

and $\tilde{\Delta}^{(n-1)}$ is defined by

$$\tilde{\Delta}^{(n-1)} = \begin{cases} \text{Id}_B & \text{if } n = 1, \\ (\tilde{\Delta}^{(n-2)} \otimes \text{Id}_B) \circ \tilde{\Delta} & \text{otherwise.} \end{cases}$$

In particular, $\phi_1 = \phi_1 \leftarrow \epsilon_\delta$.

Proposition

For any $x \in B$, with $\varepsilon_\Delta(x) = 0$,

$$\phi_1(x) = \sum_{n=1}^{\infty} \epsilon_\delta^{\otimes n} \circ \tilde{\Delta}^{(n-1)}(x) \frac{X(X-1) \cdots (X-n+1)}{n!}.$$

For any non empty graph $G \in \mathcal{H}_G$,

$$\begin{aligned}
 \phi_1(G) &= \sum_{n=1}^{\infty} \sum_{\substack{V(G)=I_1 \sqcup \dots \sqcup I_n, \\ I_1, \dots, I_n \neq \emptyset}} \epsilon_{\delta}(G|_{I_1}) \cdots \epsilon_{\delta}(G|_{I_n}) \frac{X(X-1) \cdots (X-n+1)}{n!} \\
 &= \sum_{n=1}^{\infty} \sum_{\substack{V(G)=I_1 \sqcup \dots \sqcup I_n, \\ I_1, \dots, I_n \neq \emptyset, \\ E(G|_{I_1}) = \dots = E(G|_{I_n}) = \emptyset}} \frac{X(X-1) \cdots (X-n+1)}{n!}.
 \end{aligned}$$

Let G be a graph.

- A n -coloration of G is a map from $V(G)$ to $\{1, \dots, n\}$.
- A n -coloration is valid if any two neighbors in G have different colors.

The chromatic polynomial of G is defined by

$$\forall n \in \mathbb{N}, \quad P_{\text{chr}}(G)(n) = \#\{\text{valid } n\text{-colorations of } G\}.$$

Theorem

$P_{\text{chr}} : \mathcal{H}_{\mathcal{G}} \longrightarrow \mathbb{C}[X]$ is the morphism ϕ_1 .

Consequently,

$$P_{\text{chr}}(G)(X + Y) = \sum_{V(G)=I \sqcup J} P_{\text{chr}}(G|_I)(X) P_{\text{chr}}(G|_J)(Y),$$

$$P_{\text{chr}}(G)(0) = \delta_{G,1},$$

$$P_{\text{chr}}(G)(1) = \begin{cases} 1 & \text{if } E(G) = \emptyset, \\ 0 & \text{otherwise.} \end{cases}$$

These characterize the chromatic polynomial, and imply that

$$P_{\text{chr}}(G)(XY) = \sum_{\sim} P_{\text{chr}}(G/\sim)(X) P_{\text{chr}}(G|\sim)(Y).$$

Antipode

Let (B, m, Δ, δ) be a double bialgebra. We assume that the counit ϵ_δ has an inverse α in the monoid of characters of (B, m, Δ) . Then (B, m, Δ) is a Hopf algebra, of antipode

$$S = (\alpha \otimes \text{Id}) \circ \delta.$$

As a consequence, if (B, m, Δ, δ) is a double bialgebra such that (B, m, Δ) is a Hopf algebra, then (B, m) is commutative.

Link with morphisms to $\mathbb{C}[X]$

Let $\phi_1 : B \longrightarrow \mathbb{C}[X]$ be a double bialgebra morphism. Then ϵ_δ has an inverse α in the monoid of characters of (B, m, Δ) , given by

$$\alpha(a) = \phi_1(a)(-1).$$

Moreover, (B, m, Δ) is a Hopf algebra, and its antipode is given by

$$S(x) = \underbrace{(\phi_1 \otimes \text{Id}) \circ \delta(x)}_{\in \mathbb{C}[X] \otimes B} \Big|_{X=-1}.$$

Question

What is $P_{\text{chr}}(G)(-1)$?

Using Stanley's result, or a double bialgebra morphism from graphs to oriented graphs, we obtain

Theorem

$$P_{\text{chr}}(G)(-1) = (-1)^{|V(G)|} \#\{G' \text{ acyclic orientation of } G\}.$$

Antipode of $\mathcal{H}_{\mathcal{G}}$

$(\mathcal{H}_{\mathcal{G}}, m, \Delta)$ is a Hopf algebra. Its antipode is given by

$$\begin{aligned} S(G) &= \sum_{\sim} P_{\text{chr}}(G/\sim)(-1)^{|G/\sim|} G|\sim \\ &= \sum_{\sim} (-1)^{|G/\sim|} \#\{\text{acyclic orientations of } G/\sim\} G|\sim. \end{aligned}$$

This formula was proved by Benedetti, Bergeron and Machacek in 2019 with combinatorial methods and a Möbius inversion.

Question

What about the Tutte polynomial?

The Fortuin and Kasteleyn's random cluster model is an algebra map $Z : \mathcal{H}_G \longrightarrow \mathbb{C}[X, Y]$, defined by

$$Z_G(X, Y) = \sum_{F \subseteq E(G)} X^{\text{cc}(G|_F)} Y^{|F|},$$

where for any $F \subseteq E(G)$, $G|_F$ is the graph defined by

$$V(G|_F) = V(G), \quad E(G|_F) = F.$$

It is related to the Tutte polynomial $T_G(X, Y)$:

$$T_G(X, Y) = (X - 1)^{-\text{cc}(G)} (Y - 1)^{-|V(G)|} Z_G((X - 1)(Y - 1), Y - 1),$$

$$Z_G(X, Y) = X^{\text{cc}(G)} Y^{|V(G)| - \text{cc}(G)} T_G\left(\frac{X}{Y} + 1, Y + 1\right).$$

Proposition

The following map is a Hopf algebra morphism:

$$\phi_0 : \begin{cases} (\mathcal{H}_G, m, \Delta) & \longrightarrow & (\mathbb{C}[X], m, \Delta) \\ G & \longmapsto & X^{|V(G)|}. \end{cases}$$

For any $y \in \mathbb{C}$, the following map is a character:

$$\lambda_y : \begin{cases} \mathcal{H}_G & \longrightarrow & \mathbb{C} \\ G & \longmapsto & \sum_{H \in \text{Cov}(G)} y^{|E(H)|}, \end{cases}$$

where $\text{Cov}(G)$ is the set of covering subgraphs of G .

Definition

Let G a graph and H a subgraph of G .

- ① H is a spanning subgraph of G if $V(H) = V(G)$.
- ② H is a covering subgraph of G if $V(H) = V(G)$, and G and H have the same connected components.

Theorem

For any graph G , $Z_G(X, y) = (\phi_0 \otimes \lambda_y) \circ \delta(G)$. In other words,

$$Z_\bullet(X, y) = \phi_0 \rightsquigarrow \lambda_y.$$

Consequently, for any $y \in \mathbb{C}$,

$Z_\bullet(X, y) : (\mathcal{H}_G, m, \Delta) \longrightarrow (\mathbb{C}[X], m, \Delta)$ is a Hopf algebra morphism.

In other words,

$$Z_G(X_1 + X_2, Y) = \sum_{V(G)=I \sqcup J} Z_{G|_I}(X_1, Y) Z_{G|_J}(X_2, Y).$$

Moreover, there exists a unique character μ_y such that

$$Z_\bullet(X, y) = P_{\text{chr}} \rightsquigarrow \mu_y.$$

The character μ_y is given by

$$\mu_y(G) = Z_G(1, y).$$

Proposition

For any graph G ,

$$\mu_y(G) = Z_G(1, y) = (1 + y)^{|E(G)|}.$$

Direct combinatorial consequences:

$$Z_G(X, 0) = \phi_0(G), \quad Z_G(X, -1) = \text{chr}_G(X).$$

We can also obtain the classical interpretations of $T_G(1, 2)$, $T_G(1, 1)$, $T_G(2, 1)$ (counting covering subgraphs, covering subforests, spanning subforests of G), as well as combinatorial interpretations of $Z_G(x, y)$ in terms of colourings of vertices and edges.

Similar double bialgebra structures can be defined:

- on mixed graphs (or partially oriented graphs).
Subobjects: graphs, oriented graphs.
Quotient: acyclic mixed graphs.
Subquotient: acyclic oriented graphs, posets.
- on hypergraphs.
Subobjects: graphs.
- On noncrossing partitions.
- On families of boolean functions (including rank maps of matroids).
- ...

This gives other polynomial invariants.

It is possible to define double bialgebraic structures on species.

Applying the bosonic Fock functor, we recover the preceding objects.

Applying the full Fock functor, we obtain noncommutative versions of these objects, which are no more double bialgebras by lack of commutativity, with similar morphisms between them.

ArXiv references

Bialgebras in cointeraction, the antipode and the eulerian idempotent. arXiv:2201.11974

The Fortuin-Kasteleyn polynomial as a bialgebra morphism and applications to the Tutte polynomial. With Claudia Malvenuto. arXiv:2404.05497

Thank you for your attention!